EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Pollution; Types of pollution

1. Which one is not the correct statement ?
Pollutions means :
(A) The presence of anything in the environment in excess of the required limit
(B) Anything put into the environment which was not there in its natural state.
(C) Direct or indirect changes in one or more components of the ecosystem which are not harmful to living organisms
(D) Release of any chemical substance in the environment with harmful effects

CEV001

Atmospheric pollution : Tropospheric pollution (Gaseous Air Pollution; Particulate Pollutants)

2. Greenhouse effect is related to -
(A) Cultivation of green plants
(B) Cultivation of vegetables in houses
(C) Global warning
(D) green algae

CEV004

3. Which option is correct for dust.
(A) Dust is composed of fine solid particles
(B) It is produced during crushing, grinding and attribution of solid materials
(C) Sand from sand blasting and saw dust from wood works
(D) All of the above

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CEV071

4. The acid rain possesses :
(A) Sulphuric acid (B) nitric acid (C) sulphurous acid (D) all of these

CEV002

5. Lung diseases are four times more in urban areas than rural areas. This is due to the presence of :-
(A) SO_2 (B) CO_2 (C) N_2 (D) Water-vapour

CEV003

6. Acid rains are produced by-
(A) Excess production of NH_3 by industry and coal gas
(B) Excess release of carbon monoxide by incomplete combustion
(C) Excess formation of CO_2 by combustion and animal respiration
(D) Excess NO_2 and SO_2, from burning of fossil fuels

CEV005

7. If there was no CO_2 in the earth's atmosphere, the temperature of earth's surface would be -
(A) Less than the present
(B) The same
(C) Dependent on the amount of oxygen in the atmosphere
(D) Higher than the present

CEV006

8. Taj mahal is threatened due to effect of :-
(A) Chlorine
(B) Sulphur dioxide
(C) Oxygen
(D) Hydrogen

CEV011

For More Join: @IITJEE_Advanced

9. Among the following, the one that is not a greenhouse gas is :-
(A) Nitrous oxide (B) Methane (C) Ozone (D) Sulphur dioxide

CEV019

10. Which of the following is a sink for CO ?
(A) Haemoglobin (B) Micro-organisms present in the soil
(C) Oceans (D) Plants

CEV020

Smog (Classical Smog and Photochemical Smog)

11. For smog, which option is correct.
(A) Smog are of two types – classical smog and photochemical smog.
(B) It is nonviable particulate pollutant.
(C) Photochemical smog are O_3, NO, acrolein, formaldehyde and peroxyacetyl nitrate.
(D) All of the above

CEV072

12. Pick up the correct statement
(A) CO which is major pollutant resulting from the combustion of fuels in automobiles plays a major role in photochemical smog
(B) Classical smog has an oxidizing character while the photochemical smog is reducing in character
(C) Photochemical smog occurs in daytime whereas the classical smog occurs in early morning hours
(D) During formation of smog the level of ozone in the atmosphere goes down

CEV008

Atmospheric pollution: Stratospheric pollution (Formation and Breakdown of Ozone, Ozone Hole, Adverse Effect of Depletion of Ozone Layer)

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13. CF_2Cl_2 is responsible for the decomposition of ozone to form oxygen. Which of the following reacts with ozone to form oxygen?
(A) Cl_2 (B) $Cl^{\ominus}$ (C) $F^{\ominus}$ (D) $\dot{C}l$

CEV007

14. P. A.N, stands for-
(A) Peroxyacetyl nitrite (B) Peroxyacetyl nitrate
(C) peroxyacetyl nitrile (D) Pyridineaceto-nitrite

CEV009

15. In coming years, skin related disorders will become more common due to -
(A) Air pollution (B) Excessive use of detergents
(C) Depletion of ozone layer (D) Water pollution

CEV012

Water pollution; Causes of water pollution (Pathogens, Organic, Wastes, Chemical Pollutants)

16. Which one is not a water pollutant ?
(A) Automobile exhaust (B) Plant nutrients
(C) Oxygen demanding wastes (D) Disease causing agents

CEV013

17. Major source of methane in India is :
(A) fruit orchards (B) sugar cane plantation
(C) rice fields (D) wheat fields

CEV018

For More Join: @IITJEE_Advanced

Biochemical Oxygen Demand(BOD); Eutrophication

18. Choose the correct option.
(A) Clean water would have BOD value less than 5 ppm where as highly polluted Water could have BOD value 17 ppm.
(B) Acids and salts are water soluble chemical pollutants.
(C) Eutrophication is beneficial for plants but harmful for animals of Water.
(D) All of the above

CEV073

19. When huge amount of sewage is dumped in a river, the BOD will :-
(A) Increase (B) Remain unchanged
(C) Decrease (D) Increase or decrease

CEV014

20. Eutrophication is caused by
(A) Increase nutrient concentration (B) Petrochemical and fertilizer plant effluents
(C) $NO_3^{1\ominus}$ and $SO_4^{2\ominus}$ present in acid rains (D) Mine effluents

CEV016

International standards for drinking water (Fluoride, Lead, Sulphate, Nitrate, Iron, Manganese, Aluminium, Copper, Zinc, Cadmium)

21. Maximum prescribed concentration of Fe and Mn in drinking water respectively.
(A) 0.2 and 0.05 (B) 0.2 and 3.0 (C) 5.0 and 0.05 (D) 0.02 and 0.05

CEV074

22. Excess of nitrate in drinking water causes.
(A) Blue baby syndrome (B) Laxative
(C) Tooth decay (D) None of the above

CEV075

23. A dental disease characterised by mottling of teeth is due to presence of a certain chemical element in drinking water. Which is the element -
(A) Fluorine (B) Mercury (C) Boron (D) Chlorine

CEV015

Soil pollution; Pesticides

24. Indiscriminate use of DDT is undesirable because -
(A) It is harmful (B) It is degradable
(C) It causes mutation (D) It is accumulated in food chain

CEV017

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Smog (Classical Smog and Photochemical Smog)

1. The Smog is essentially caused by absence of

(A) O_3 (B) N_2 (C) Cl_2 (D) oxides of S and N

CEV106

Atmospheric pollution: Stratospheric pollution (Formation and Breakdown of Ozone, Ozone Hole, Adverse Effect of Depletion of Ozone Layer)

2. The pollutant which is/are not responsible for ozone layer depletion.

(A) SO_2 (B) DDT (C) CO_2 (D) Freons

CEV107

Water pollution; Causes of water pollution (Pathogens, Organic, Wastes, Chemical Pollutants)

3. Which option is/are correct.

(A) The recommended concentration of fluoride ion is 1ppm.

(B) Less than 1 ppm is cause of teeth decay

(C) more than 1 ppm is cause of brown mottling of teeth.

(D) None of the above

CEV108

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COMPREHENSION TYPE QUESTIONS

Atmospheric pollution: Stratospheric pollution (Formation and Breakdown of Ozone, Ozone Hole, Adverse Effect of Depletion of Ozone Layer)

Paragraph for Question No. 4 to 5

In summer season, nitrogen dioxide and methane react with chlorine monoxide (reaction i) and chlorine atoms (reaction ii) forming chlorine sinks, preventing much ozone depletion, whereas in winter, special type of clouds called polar stratospheric clouds provide surface on which chlorine nitrate formed (in reaction i) gets hydrolysed to form hypochlorous acid (reaction (iii)). It also reacts with hydrogen chloride produced as per reaction (ii) to give molecular chlorine.

$\dot{Cl}O(g) + NO_2(g) \rightarrow ClONO_2(g)$ (i)

$\dot{Cl}(g) + CH_4(g) \rightarrow \dot{C}H_3(g) + HCl(g)$ (ii)

$ClONO_2(g) + H_2O(g) \rightarrow HOCl(g) + HNO_3(g)$ (iii)

$ClONO_2(g) + HCl(g) \rightarrow Cl_2(g) + HNO_3(g)$ (iv)

When sunlight returns to the Antarctica in the spring, the Sun's warmth breaks up the clouds and HOCl and Cl_2 are photolysed by sunlight, as given in reactions (v) and (vi).

$HOCl(g) \xrightarrow{h\nu} \dot{O}H(g) + \dot{Cl}(g)$ (v)

$Cl_2(g) \xrightarrow{h\nu} 2\dot{Cl}(g)$ (vi)

The chlorine radicals thus formed, initiate the chain reaction for ozone depletion.

For More Join: @IITJEE_Advanced

4. Which option is correct for ozone depletion.
(1) In Summer season, Chlorine sinks formation prevents ozone Layer depletion.
(2) In winter season, it forms polar stratospheric cloud
(3) In spring season, ozone layer depletion starts.
(4) All of the above

CEV109

5. Which of the following species are responsible for ozone depletion

(1) $\dot{Cl}O$ (2) $\dot{C}l$ (3) Both A and B (4) None of these

CEV110

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EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Atmospheric pollution : Tropospheric pollution (Gaseous Air Pollution ; Particulate Pollutants)

1. The Sulphur compound which is released by scooter and cars as pollutant. find the value of x for that compound.

A = Number of σ-bond

B = Number of pπ-pπ bonds

C = Number of pπ-dπ bonds

D = Number of lone pair on central atom.

x = A + B + C + D.

CEV022

2. Gaseous pollutant of carbon are x and y. Find the number of sums of σ and π bond in x and y is.

CEV025

3. In acid rain

$2SO_{2(g)} + O_{2(g)} + 2H_2O_{(l)} \rightarrow \underset{(x)}{2H_2SO_{4(aq)}}$

$4NO_{2(g)} + O_{2(g)} + 2H_2O_{(l)} \rightarrow \underset{(y)}{4HNO_{3(aq)}}$

Rankers Academy JEE

Find the sum of oxidation state of x & y is ? Mark your answer as sum of digit

CEV026

4. In the following reaction

$NO + O_3 \longrightarrow \underset{(\text{Brown gas})}{x} + O_2$

What is the difference between bond angle of x and O_3. (Mark your answer as sum of digits).

CEV029

Atmospheric pollution: Stratospheric pollution (Formation and Breakdown of Ozone, Ozone Hole, Adverse Effect of Depletion of Ozone Layer)

5. Total number of lone pairs in chloro–floro carbon mark your answer as sum of digit

CEV027

6. Find the number of chief pollutants which are responsible for ozone depletion.

SO_2, CO_2, CO, NO_2, CH_4, CFC, Cl_2, Br_2, I_2, N_2, He, Ne

CEV030

Biochemical Oxygen Demand (BOD); Eutrophication

7. Clean water has BOD value less than – ppm.

CEV023

International standards for drinking water (Fluoride, Lead, Sulphate, Nitrate, Iron, Manganese, Aluminium, Copper, Zinc, Cadmium)

8. Maximum prescribed concentration of cadmium in drinking water is 'x'. Find the value of (x × 1000).

CEV024

For More Join: @IITJEE_Advanced

Green Chemistry

9. Tetrachloro ethene is used as solvent for drycleaning How many total numbers of σ-bonds + pπ – pπ bonds + pπ – dπ bonds.

CEV028

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

International standards for drinking water (Fluoride, Lead, Sulphate, Nitrate, Iron, Manganese, Aluminium, Copper, Zinc, Cadmium)

10. Maximum prescribed concentration of Aluminium in drinking water is …….

CEV111

11. What is the bond order of the specie which is the main air pollutant among CO_2, CO, N_2 and S?

CEV112

12. What is the prescribed concentration of the metal in drinking water according to international standards which shows the highest oxidation number among given metals –

Metal	Maximum concentration (ppm)
Fe, Al	0.2
Mn	0.05
Cu	3.0
Zn	5.0
Cd	0.005
Pb	50

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CEV113

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Correct statement is : **[JEE (Main) 2023]**
(1) An average human being consumes more food than air
(2) An average human being consumes nearly 15 times more air than food
(3) An average human being consumes equal amount of food and air
(4) An average human being consumes 100 times more air than food

CEV114

2. **Assertion :** Nitrogen and Oxygen are the main components in the atmosphere but these do not react to form oxides of nitrogen.
Reason : The reaction between nitrogen and oxygen requires high temperature.
[JEE (Main) 2015]
(1) Both assertion and reason are correct, and the reason is the correct explanation for the assertion
(2) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion
(3) The assertion is incorrect, but the reason is correct
(4) Both the assertion and reason are incorrect

CEV037

3. The gas leaked from a storage tank of the Union Carbide plant in Bhopal gas tragedy was :
[JEE (Main) 2013]
(1) Methylisocyanate (2) Methylamine (3) Ammonia (4) Phosgene

CEV036

4. When rain is accompanied by a thunderstorm, the collected rain water will have a pH value
[AIEEE 2003]
(1) Slightly higher than that when the thunderstorm is not there
(2) Uninfluenced by occurrence of thunderstorm
(3) pH depends on the amount of dust in air
(4) Slightly lower than that of rain water without thunderstorm

Pradeep Academy JEE

CEV031

5. How can photochemical smog be controlled? **[JEE (Main) 2023]**
(1) By using tall chimneys
(2) By complete combustion of fuel
(3) By using catalytic converters in the automobiles/industry
(4) By using catalyst

CEV115

6. Correct statement about smog is **[JEE (Main) 2023]**
(1) NO_2 is present in classical smog
(2) Both NO_2 and SO_2 are present in classical smog
(3) Photochemical smog has high concentration of oxidizing agents
(4) Classical smog also has high concentration of oxidizing agents

CEV049

7. The smog is essentially caused by the presence of : **[AIEEE 2004]**
(1) O_2 and O_3 (2) O_3 and N_2
(3) Oxides of sulphur and nitrogen (4) O_2 and N_2

CEV032

8. Identify X, Y and Z in the following reaction. (Equation not balanced)

$$Cl\dot{O} + NO_2 \rightarrow \underline{X} \xrightarrow{H_2O} \underline{Y} + \underline{Z}$$ **[JEE (Main) 2023]**

(1) X=$ClONO_2$, Y=HOCl, Z = NO_2 (2) X=$ClNO_2$, Y=HCl, Z=HNO_3
(3) X=$ClONO_2$, Y=HOCl, Z=HNO_3 (4) X=$ClNO_3$, Y=Cl_2, Z=NO_2

CEV050

For More Join: @IITJEE_Advanced

9. $\dot{Cl} + CH_4 \rightarrow A + B$

A and B in the above atmospheric reaction step are **[JEE (Main) 2022]**

(1) C_2H_6 and Cl_2 (2) $\dot{C}HCl_2$ and H_2

(3) $\dot{C}H_3$ and HCl (4) C_2H_6 and HCl

CEV047

10. The reaction that is NOT involved in the ozone layer depletion mechanism in the stratosphere is: **[JEE (Main) 2019]**

(1) $HOCl(g) \xrightarrow{h\nu} \dot{O}H(g) + \dot{C}l(g)$ (2) $CF_2Cl_2(g) \xrightarrow{uv} \dot{C}l(g) + \dot{C}F_2Cl(g)$

(3) $CH_4 + 2O_3 \rightarrow 3CH_2 = O + 3H_2O$ (4) $\dot{C}lO(g) + O(g) \longrightarrow \dot{C}l(g) + O_2(g)$

CEV042

11. The upper stratosphere consisting of the ozone layer protects us from the sun's radiation that falls in the wavelength region of : **[JEE (Main) 2019]**

(1) 600-750 nm (2) 0.8-1.5 nm

(3) 400-550 nm (4) 200-315 nm

CEV043

12. Identify the wrong statements in the following : **[AIEEE 2008]**

(1) Chlorofluorocarbons are responsible for ozone layer depletion

(2) Greenhouse effect is responsible for global warming

(3) Ozone layer does not permit infrared radiation from the sun to reach the earth

(4) Acid rains are mostly because of oxides of nitrogen and sulphur

CEV034

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13. Biochemical Oxygen Demand (BOD) is the amount of oxygen required (in ppm): **[JEE (Main) 2020]**

(1) by anaerobic bacteria to breakdown inorganic waste present in a water body.

(2) for the photochemical breakdown of waste present in 1 m^3 volume of a water body.

(3) by bacteria to break-down organic waste in a certain volume of a water sample.

(4) for sustaining life in a water body.

CEV044

14. Biochemical Oxygen Demand (BOD) value can be a measure of water pollution caused by the organic matter. Which of the following statements is correct - **[JEE (Main) 2018]**

(1) Anaerobic bacteria increase the BOD value

(2) Aerobic bacteria decrease the BOD value

(3) Polluted water has BOD value higher than 10 ppm

(4) Clean water has BOD value higher than 10 ppm

CEV040

15. A water sample has ppm level concentration of the following metals: Fe= 0.2; Mn = 5.0; Cu = 3.0; Zn = 5.0. The metal that makes the water sample unsuitable for drinking is : **[JEE (Main) 2019]**

(1) Zn (2) Fe (3) Mn (4) Cu

CEV041

16. A water sample has ppm level concentration of following anions $F^{\ominus}$ = 10; $SO_4^{2\ominus}$ = 100; $NO_3^{\ominus}$ = 50 the anion/anions that make/makes the water sample unsuitable for drinking is / are : **[JEE (Main) 2017]**

(1) only $NO_3^{\ominus}$ (2) both $SO_4^{2\ominus}$ and $NO_3^{\ominus}$

(3) only $F^{\ominus}$ (4) only $SO_4^{2\ominus}$

CEV039

For More Join: @IITJEE_Advanced

17. The concentration of fluoride, lead, nitrate and iron in a water sample from an underground lake was found to be 1000 ppb, 40 ppb, 100 ppm and 0.2 ppm, respectively. This water is unsuitable for drinking due to high concentration of : **[JEE (Main) 2016]**

(1) Fluoride (2) Lead (3) Nitrate (4) Iron

CEV038

18. Regular use of which of the following fertilizers increases the acidity of soil ? **[AIEEE 2007]**

(1) potassium nitrate (2) Urea
(3) Superphosphate of lime (4) Ammonium sulphate

CEV033

19. Match List-I with List-II

List-I **Industry**	**List-II** **Waste Generated**
(A) Steel plants	(I) Gypsum
(B) Thermal power plants	(II) Fly ash
(C) Fertilizer industries	(III) Slag
(D) Paper mills	(IV) Bio-degradable Wastes

Choose the correct answer from the options given below:

[JEE (Main) 2023]

(1) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
(2) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
(3) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
(4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

CEV116

20. Given below are two statements:

Statement I : The non bio-degradable fly ash and slag from steel industry can be used by cement industry.

Statement II : The fuel obtained from plastic waste is lead free.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

[JEE (Main) 2022]

(1) Both **Statement I** and **Statement II** are correct
(2) Both **Statement I** and **Statement II** are incorrect
(3) **Statement I** is correct but **Statement II** is incorrect
(4) **Statement I** is incorrect but **Statement II** is correct

CEV048

21. What is DDT among the following : **[JEE (Main) 2012]**

(1) Greenhouse gas (2) A fertilizer
(3) Biodegradable pollutant (4) Non-biodegradable pollutant

CEV035

22. Green chemistry in day–to–day life is in the use of: **[JEE (Main) 2021]**

(1) Chlorine for bleaching of paper
(2) Large amount of water alone for washing clothes
(3) Tetrachloroethene for laundry
(4) Liquified CO_2 for dry cleaning of clothes

CEV045

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	D	D	A	D	A	B	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	C	D	B	C	A	C	D	A	A
Que.	21	22	23	24						
Ans.	A	A	A	D						

EXERCISE (O-2)

Que.	1	2	3	4	5
Ans.	A,B,C	A,B,C	A,B,C	4	3

EXERCISE (S)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	5	7	2	8	3	2	5	5	6	0.20
Que.	11	12								
Ans.	3.00	0.05								

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	1	4	3	3	3	3	3	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	3	2	3	3	3	4	1	1
Que.	21	22								
Ans.	4	4								

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